VEERAJ PANCHAL

Navi Mumbai 8898933595 veerajkpanchal@gmail.com
GitHub LinkedIn

Education

D. J. Sanghvi College of Engineering, Mumbai *Expected 2028*
B.Tech in Artificial Intelligence & Machine Learning

Bharati Vidyapeeth Institute of Technology, Navi Mumbai *Feb 2025*
Diploma in Computer Technology **Aggregate: 92.3%**

Experience

Software Developer Intern | Raiseon Automation *Aug 2024*

- Developed a C# desktop application integrating the Cognex DataMan 250 SDK for automated barcode inspection.
- Implemented hardware-triggered image acquisition, barcode capture, and automated inspection workflows.

Projects

ReconAI – AI-Assisted Financial Reconciliation Platform *May 2026 – Present*
GitHub

- Developed an AI-assisted financial reconciliation platform using Django, Pandas, Next.js and Sentence Transformers for multi-source transaction data.
- Implemented semantic column mapping with **all-MiniLM-L6-v2** to map unfamiliar financial schemas to canonical fields.
- Built a deterministic reconciliation engine handling transaction matching, date tolerance, amount discrepancies, duplicates, sign reversals and grouped transactions.
- Added exception intelligence using **KNN, Logistic Regression, Isotonic Regression and Apriori** for pattern analysis and confidence calibration.
- Added database persistence for reconciliation runs and results; benchmarked end-to-end processing up to **1.3 lakh entries**.
- Achieved $\leq$**9s** for 10K-entry datasets, with larger runs completing in approximately 1 minute.

AI Smart Traffic Management System *Jan 2026 – Present*
GitHub

- Developing an AI-based traffic management system using computer vision for real-time vehicle detection and traffic analysis.
- Integrated YOLOv8 and OpenCV for vehicle detection and traffic-density estimation from CCTV/video feeds.
- Designed adaptive traffic-signal logic to dynamically adjust signal duration based on detected traffic density.

Grievance Management System *Mar 2026*
GitHub

- Developed a full-stack grievance management platform with role-based access, complaint tracking and resolution workflows.

Dynamic Result Analyzer *Mar 2025*
GitHub

- Developed a result-analysis application using **Spring Boot and Selenium** for automated student performance evaluation and academic insights.

Technical Skills

Languages: Java, Python, JavaScript, SQL
Frameworks & Libraries: Django, Next.js, React, Pandas, OpenCV, Sentence Transformers, Scikit-learn
AI/ML: NLP, Embeddings, Semantic Search, RAG, YOLOv8
ML Techniques: KNN, Logistic Regression, Isotonic Regression, Apriori, Feature Engineering, Similarity Scoring, Model Calibration
Databases: PostgreSQL, MySQL, Prisma ORM
Core CS: DSA, Algorithms, OOP, DBMS, OS, CN
Backend: REST APIs, Authentication, MVC Architecture
Tools: Git, GitHub, Selenium, Android Studio